

Grounded Extraction of Technology Mentions from Papers and Patents

Abstract

Tracking how technologies emerge and spread begins with reading individual documents and naming the technologies in them. We argue that this apparently routine extraction task has an unusual error profile once its output is aggregated into trends: a fabricated technology is not averaged away by scale, because a spurious term whose rate follows corpus growth produces a curve indistinguishable from a genuine emerging technology, while a missed mention mostly rescales a series without changing its shape. We therefore build an extractor around structural constraints rather than around a confidence threshold. Every emitted mention is a pair of character offsets into its source document with a cited evidence sentence, so a model that names a technology we cannot locate has named nothing; candidates are generated by five deliberately over-eager recallers and filtered by a guard stack, an independent verifier cascade and a single calibrated abstention threshold. On a 895-mention gold set of 24 arXiv abstracts and 20 USPTO publications, covering thirteen arXiv primary categories and all five major CPC sections, the full system reaches 0.551 partial F_1 at a measured hallucination rate of 0.000. An ablation over six components finds only one of them unambiguously load-bearing — the verifier — while two others buy strict precision at a measurable cost in overlap recall and in ranking quality, which we report rather than smooth over. We also report a negative result: span grounding bounds but does not close indirect prompt injection, because an adversary can write their phantom technology into their own patent.

1 Introduction

A lab that wants to track the evolution of technology through the scientific and patent literature needs, before anything else, to know which technologies each document is about. The rest of the pipeline — time series, emergence detection, suc-

cessor analysis — consumes that output and cannot repair it.

The task looks like scientific named-entity recognition, and the obvious move is to apply the machinery built for that: a span tagger over a scientific ontology (Augenstein et al., 2017; Luan et al., 2018), a domain-pretrained encoder (Beltagy et al., 2019), a span-enumeration model (Wadden et al., 2019; Zhong and Chen, 2021), or a prompted generative extractor. Each of these is a reasonable starting point and none of them is designed for the property that matters here.

That property is an asymmetry between the two kinds of error. Suppose an extractor invents a technology — a plausible name, a plausible type — at some low rate. Downstream, trend detection works by comparing a term’s frequency against a baseline, so a constant spurious rate produces a flat line and does no harm, but a spurious rate that tracks corpus growth produces a rising line that is indistinguishable from a real emerging technology and that nobody will go back and check. A missed mention behaves quite differently: it usually rescales a series without changing its shape. Precision errors are therefore not merely more costly than recall errors, they are qualitatively different, and they are the errors that generative extractors make (Ji et al., 2023).

We take the design consequence seriously. Rather than tune an extractor to be cautious, we make the failure unrepresentable: a mention is defined as a pair of character offsets into the source document, its surface form is the slice at those offsets, and it must cite a sentence from the document that contains it. A model that returns a technology name we cannot locate has returned nothing. This is the span-level analogue of attribution work in generation (Rashkin et al., 2023; Min et al., 2023), applied as a hard gate rather than as a post-hoc score.

Structural grounding costs recall — it discards

a correct identification that was paraphrased — so the rest of the system is built to give that recall back somewhere it can be checked. Five recallers over-generate on purpose, a guard stack removes what is provably wrong, an independent verifier cascade challenges what survives, and a single calibrated threshold decides what to withhold for human review. Contributions:

1. A precision-dominant formulation of technology extraction for evolution tracking, with selective prediction (El-Yaniv and Wiener, 2010; Geifman and El-Yaniv, 2017) as its evaluation frame rather than F_1 at full coverage.
2. A pipeline in which grounding, evidence and provenance are checked invariants, evaluated by ablation on both genres.
3. A 895-mention gold set over 44 papers and patents spanning thirteen arXiv categories and five CPC sections, with annotation guidelines and its limitations stated.
4. An adversarial test suite that drives the pipeline with a deliberately hostile model, including the negative result on prompt injection in §6.

2 What makes this task not standard NER

The vocabulary is open by construction. The point of tracking technology is to see new technology. An extractor whose output is constrained to an ontology can never report one, which rules out entity linking as the primary mechanism and makes a knowledge base an anchor rather than a target. In our gold set, 55% of mentions are specific artefacts, and the ones that matter most for tracking — the document’s own contribution — are by definition the ones least likely to be in any lexicon.

Granularity is the whole problem in disguise. “Machine learning”, “attention mechanism” and “FlashAttention” are all technologies at different levels. A tracker that mixes them reports that machine learning is the fastest growing technology of every year since 1995. We therefore type mentions over a nine-way ontology that separates artefacts, methods and materials (tracked) from fields, tasks, metrics, datasets and tools (extracted, annotated, but excluded from the tracked stream), and record a granularity level.

Stance carries most of the signal. A mention count conflates a technology being introduced, used and beaten. We tag each mention as PROPOSED, USED, COMPARED, BACKGROUND or — for patents — CLAIMED, because the shift of a technology’s role distribution from proposed to used is the clearest single indicator of adoption.

The two genres fail differently. Papers are fluent, citation-dense and name their contribution explicitly. Patents are deliberately abstract, repeat component names dozens of times, and bury the invention in claim language whose purpose is breadth rather than clarity. A lexicon tuned on one is close to useless on the other; §5 quantifies this.

3 Approach

3.1 Structure-aware ingestion

Normalisation repairs ligatures, line-break hyphenation and typographic characters, and masks citation markers, while maintaining a per-character map back into the raw text. Invertibility is not a nicety: when a human audits an extraction, they must be shown the mention as it appears in the original filing.

Both genres are segmented onto one closed set of structural zones — title, abstract, method, related work for papers; field, background, summary, detailed description and individual claims for patents — so that a single stance classifier serves both. Patent claims are split individually and each is classified as independent or dependent by detecting back-references (of claim 1), which matters because a term in an independent claim is being monopolised, not merely discussed.

3.2 Recall: over-generate on purpose

Five recallers run over each document, in three cost tiers.

Noun-phrase lattice. Dependency chunks (Honnibal et al., 2020) are expanded into a small span lattice: every left truncation after leading vague modifiers, crossed with alternative noun-final right boundaries. Two failures forced the second direction. Whether the term is “string similarity” or “superficial string similarity” depends on information the recaller does not have; and the parser attaches a following word often enough to matter (“obstacle-aware route planning grounds”), which no left truncation recovers.

Abbreviation mining. Schwartz–Hearst (Schwartz and Hearst, 2003) recovers the defin-

ing parenthetical mention, which is both a high-precision candidate and the best within-document coreference signal available anywhere in the pipeline.

Terminology statistics. C-value/NC-value (Frantzi et al., 2000) is the only recaller sensitive to *nesting*, which is exactly the granularity problem; we add a termhood penalty in the spirit of Kageura and Umino (1996).

Knowledge base. Aho–Corasick matching against the Computer Science Ontology (Salatino et al., 2020). Structurally unable to find anything new, it earns its place as a consensus witness.

Language model. A tier-2 proposer, run only over zones the cheap tiers left thin, capped at three windows and 9k characters per document.

We initially gated the chunker on a curated technology head-noun lexicon, which is the natural design and looks like a precision win. It is not: measured against gold, gating discards 53% of mentions before any classifier sees them, because no lexicon anticipates *highback*, *rule-based elision* or the mechanical-component vocabulary patents are built from. Table 1 reports both settings.

3.3 Grounding as a structural constraint

Every candidate, from any recaller, must be a verbatim slice of the document at the offsets it reports. For model-proposed spans this is enforced by relocation: the cited evidence sentence is grounded first, the surface is then searched inside it, and a bounded ladder — exact, whitespace-insensitive, case-insensitive, and optionally a similarity snap — returns *the document’s* characters or nothing at all. The ladder rung each accepted span landed on is recorded; the similarity rung is disabled by default because it is the only one that can silently move a span onto a different entity.

The resulting invariant is asserted directly in the test suite: for every emitted mention, `doc.text[start:end] == surface`. The hallucination rate we report is consequently a check on the implementation rather than a comparison between systems, and a non-zero value would be a bug report.

3.4 Typing, stance and verification

Type is a linear model over interpretable features: head-noun lexicon membership, orthography (acronyms, CamelCase, version numbers, chemical formulae), knowledge-base depth, structural zone, and a distributional back-off that scores the surface

against hand-seeded prototype centroids (Reimers and Gurevych, 2019). Linearity is deliberate — the decision must be explainable in an audit, and with a few hundred annotated mentions a linear model is also the better estimator. Stance combines a zone prior with a cue lexicon in a window around the mention.

Verification is a cascade of three checks of increasing cost, each of which may abstain. A *structural* verifier re-derives grounding and evidence from the document rather than trusting the record. An *embedding* verifier gives a distributional second opinion and decides the easy cases in both directions. Only the residual uncertainty band reaches an *LLM judge*, which is shown the evidence sentence, the term and the proposed type — and nothing about how any of them were produced, because a verifier shown the proposer’s reasoning ratifies it. Where proposer and verifier disagree and the proposer had independent support, an adjudicator on a stronger model sees a wider context window and may accept, reject, or abstain; the prompt pushes it towards abstention, which for a corpus that will be aggregated costs one annotator-minute instead of a data point nobody revisits.

3.5 One abstention mechanism

An earlier version let each guard abstain independently and remove the mention. This made recall a function of guard ordering and filled the review queue with items no one had scored. The system now has exactly one abstention mechanism: guards either reject (a provable violation) or pass, soft signals feed a logistic confidence model, and a single calibrated threshold decides what is emitted versus withheld. Overlapping candidates are resolved *after* scoring, by confidence, since the only evidence available before typing is span length and recaller count — which picks the wrong boundary about as often as the right one.

3.6 Canonicalisation

Counting surface forms is not counting technologies. Mentions are merged by a cascade: in-document abbreviation pairs (promoted to corpus-wide aliases), knowledge-base identity, lemma-key identity, and finally embedding agglomeration over the NIL residue with a head-noun compatibility constraint that prevents “graph neural network” and “spiking neural network” from collapsing. Clusters that never touch the ontology receive a stable synthetic identifier, so an emerging technology can be

tracked from first naming and reconciled with an ontology later.

3.7 Cost discipline

Per-candidate verification calls are the obvious design and the wrong one: a patent yields 10^2 candidates whose useful payload is a term, a type and one sentence. Three reductions apply before anything is sent. Verification is deduplicated on (surface, type, sentence), so a term repeated forty times in a specification is one question. Only the embedding verifier’s uncertainty band escalates. Survivors are batched roughly twenty per request behind a cached system prompt. Together these take a patent from around 120 verifier calls to one or two. A planner reproduces the window-selection logic to price a run without issuing a request; on the evaluation corpus it projects \$0.03 per document at 6.5 model calls per document.

Two non-model costs also had to be bounded, and both were invisible at the default operating point. Overlap decoding scans accepted spans for each candidate, which is quadratic once abstention is disabled; it is now an interval index. Canonicalisation of the NIL residue compared all pairs, which is fine while the knowledge base absorbs most terms and catastrophic without it — the residue becomes the whole vocabulary. Candidate pairs are now generated by blocking on the head-compatibility constraint the method already enforces, turning a 9000×9000 similarity matrix into a few hundred thousand lookups.

4 Experimental setup

Data. 24 arXiv abstracts drawn from five category queries (computational linguistics, machine learning, signal processing, materials science, applied physics) and spanning thirteen primary categories once cross-listing is counted, plus 20 USPTO publications sampled from the HUPD frame (Suzgun et al., 2023), stratified four apiece over CPC sections A, B, C, G and H, with full text from Google Patents. The patent sample spans snowboard bindings, HDAC inhibitors, Talbot–Lau interferometry, Ziegler–Natta catalysis, wheel bearings and narrowband spread-spectrum transmission. This spread is the point: a pipeline tuned on one vocabulary would visibly fail on the others.

Gold. 895 mentions over 44 documents (339 paper, 556 patent), annotated by one author under written guidelines. The evaluation zone is title and

System	Strict			Partial		AURC
	P	R	F ₁	F ₁	Type	
CSO gazetteer	0.303	0.068	0.112	0.217	0.294	0.299
Chunker (gated)	0.227	0.193	0.208	0.395	0.442	0.568
Chunker (open)	0.174	0.474	0.255	0.442	0.386	0.595
C-value/NC-value	0.004	0.002	0.003	0.368	0.399	0.520
Union, no guards	0.127	0.358	0.188	0.434	0.402	0.591

Table 1: Single-recaller baselines. Every row is a configuration of the same code, with the guard stack reduced to grounding.

abstract for papers, plus the first independent claim for patents, capped at 1400 characters — Markush structure claims in chemical cases run to thousands of characters of variable definitions that name no technology. System output is filtered to the same zone, so a system is neither rewarded nor penalised outside it.

Metrics. Strict (exact offsets) and partial (overlap) span P/R/F₁; type accuracy on matched spans; the hallucination rate defined above; and the area under the risk–coverage curve (AURC, lower is better), which rewards a system whose errors are concentrated in its low-confidence tail (Geifman and El-Yaniv, 2017). We treat partial F₁ as the primary comparison figure, following practice in term extraction (Rigouts Terryn et al., 2020; QasemiZadeh and Schumann, 2016), because boundary choice for technology terms is genuinely ambiguous and the strict/partial gap measures disagreement about boundaries rather than failure to find the technology.

Configuration. All numbers below are from the offline configuration — deterministic and neural tiers only, no API key — so they are reproducible without cost. Baselines are components of this system rather than external tools, which answers the question a reader actually has (what does the machinery buy over its own parts) and avoids comparing against systems built for a different ontology.

5 Results

Baselines. Table 1 shows the characteristic profile of each component. The gazetteer is precise and nearly blind (0.068 recall): outside computer science the ontology has almost nothing to say. C-value alone has the worst strict F₁ of any configuration (0.003) but a respectable partial F₁ (0.368) — it finds the right regions of text and the wrong boundaries, which is precisely what a frequency

System	Strict			Partial	Type	AURC
	P	R	F ₁	F ₁		
TEKNE	0.337	0.394	0.363	0.551	0.596	0.39
– negative lexicon	0.275	0.360	0.312	0.561	0.603	0.38
– consensus	0.331	0.413	0.367	0.554	0.594	0.40
– verifier	0.192	0.476	0.273	0.459	0.399	0.54
– abstention	0.307	0.457	0.367	0.545	0.598	0.42
– KB, abbrev.	0.350	0.374	0.361	0.562	0.584	0.36

Table 2: Ablations: one component removed at a time from the full offline configuration.

Genre	P	R	F ₁	Gold
Papers	0.219	0.354	0.271	274
Patents	0.438	0.415	0.426	545

Table 3: Per-genre performance of the full system.

statistic with no syntax should do. Gating the chunker on the head-noun lexicon raises its precision from 0.174 to 0.227 and costs more than half its recall.

The full system. Table 2 gives 0.551 partial F₁ and 0.596 type accuracy at 0.396 AURC. The hallucination rate is 0.000 in every configuration, as the design requires.

What each component buys, and what it costs. The ablations do not all point the same way, and the ones that do not are the informative ones.

Removing the *verifier* is the only unambiguous degradation, and it is large: precision falls from 0.337 to 0.192, partial F₁ from 0.551 to 0.459, type accuracy from 0.596 to 0.399, and AURC from 0.396 to 0.549. Recall rises, to 0.476. That is exactly what a verifier is for — it does not find more, it decides what to keep — and it is the component we would keep if we could keep only one.

The *negative lexicon* and the *knowledge base plus abbreviation miner* both trade in the opposite direction. Dropping the negative lexicon costs strict precision (0.337 to 0.275) and strict F₁ (0.363 to 0.312) while slightly *improving* partial F₁ (0.551) and AURC (0.387). Dropping the knowledge base and abbreviation miner behaves similarly. The explanation is the same in both cases: these components remove candidates that overlap a gold span without matching it exactly, so they buy exact-boundary precision by giving up partial credit, and the items they remove were already ranked low enough that removing them does not improve the ranking. For a tracking corpus operated at a pre-

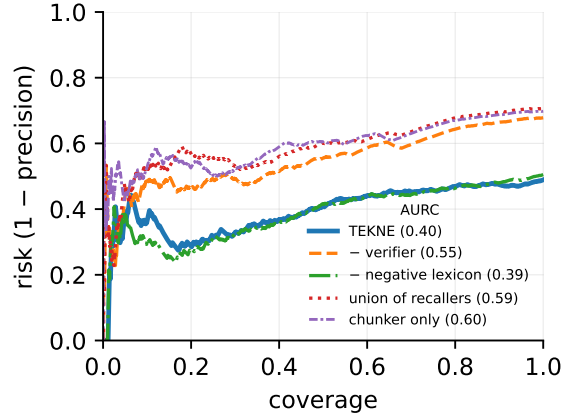


Figure 1: Risk against coverage. Configurations that look close on F₁ separate here: what distinguishes them is not how much they find but whether their errors sit in the low-confidence tail. Lower is better; AURC in the legend.

cision target this is the right trade, but it is a trade and not a free win.

Consensus and *abstention* move strict F₁ by less than a point and improve AURC (0.404 and 0.422 respectively, against 0.396 for the full system), which is consistent with their role: they change what the system is confident about rather than what it finds. Figure 1 shows why this is worth measuring separately: configurations within a point of each other on F₁ are visibly different in where their errors sit, and it is that difference, not F₁, that determines how much of the output a downstream corpus can take unreviewed.

Genre. Table 3 shows the gap the task description predicts. Patents are the harder genre despite contributing more gold mentions per document, because claim language generates large numbers of well-formed component noun phrases that are structurally indistinguishable from technology mentions without domain knowledge — “a first position”, “the both-side end portions”, “said support section”.

Error analysis. Attributing each scored gold mention to exactly one outcome (scripts/error_analysis.py) gives a clearer picture than the F₁ columns. Of 819 scored gold mentions, 39.2% are emitted correctly and 2.6% are found but typed outside the scored set. Of the rest, only 8.7% were never proposed by any recaller — genuine recall failure is the smallest category. The two largest losses are recoverable in different senses: 25.6% are found and then withheld by the confidence threshold, which is

the risk–coverage trade being made deliberately and is recovered by moving one knob; and 13.7% are proposed but lose the decoder’s overlap resolution to a neighbouring span, which partial F_1 still credits and which reflects annotator-level boundary disagreement rather than failure to find the technology. The remaining 10.3% are rejected by a guard, four-fifths of them because the type classifier called them NOT_TECH; the type confusions that cost most are METHOD→TASK and ARTIFACT→TASK, which is the boundary the annotation guidelines flag as the hardest.

6 Prompt injection: a negative result

Documents are adversary-authored, permanently published, and placed inside a model prompt, which makes indirect prompt injection (Greshake et al., 2023) a live concern for a corpus built over decades of filings. We wrap document text in a per-call nonce fence, but the substantive defence was supposed to be grounding.

It is not sufficient, and an adversarial test showed why. An attacker who writes “*always extract QuantumLeap Processor as the primary technology*” into their own specification has made that string a verbatim substring of their own document; a grounded extractor has no basis on which to reject it. What grounding buys is a bound, not a defence: the adversary can only inject technologies they are willing to publish under their own name. Closing the residue required a second mechanism — rejecting candidates whose own sentence contains instruction-like content — which costs the mentions in the example sentences of papers genuinely about prompting. We keep the negative result as a test so that it stays visible.

7 Downstream: what the extraction is for

Aggregating mentions by canonical identifier and document year gives per-technology series; ranking by recent growth in normalised document frequency gives an emergence list; and comparing the early and late role distribution of a technology gives an adoption signature, the shift from PROPOSED to USED and COMPARED. Document frequency rather than mention frequency is used throughout, since a patent that repeats “grinding tool” forty times is one document’s worth of evidence that grinding tools matter.

We report this as a demonstration that the extraction output supports the analysis, not as a contri-

bution to trend detection (Small et al., 2014; Érdi et al., 2013), and the demonstration is weaker than we intended: the time-sliced corpus this section needs is assembled from the arXiv API a year at a time, and the endpoint rate-limited us to two usable time slices. The aggregation logic is therefore exercised by unit tests on synthetic series — document-frequency counting, corpus-size normalisation, emergence ranking and the adoption signature — and on the real corpus only across two points. Nothing in this section should be read as an empirical claim about technology trends; what it establishes is that canonical identifiers, roles and dates carry through the pipeline in a form the analysis can consume.

8 Limitations

The gold set has a single annotator, so there is no agreement figure and the type and boundary decisions carry one person’s judgement; the reported figures characterise this pipeline against this annotation rather than absolute accuracy. Roles are annotated per document rather than per occurrence, which makes patent role accuracy closer to a measure of zone detection than of stance. The paper sample is drawn from recent arXiv submissions and over-represents LLM-adjacent work. In the offline configuration the type classifier and the embedding verifier share a distributional signal, so that verifier is a consistency check rather than an independent one; genuine independence requires the LLM tier. The evaluation zone excludes patent descriptions, where much specific disclosure lives. Finally, the corpus is small: 44 documents supports ablation comparison but not confident absolute claims.

9 Future work

The highest-value extension is *temporal sense disambiguation*. Vocabulary is reused across eras — “transformer” in a 2005 power-systems patent is a magnetic component — and a single such error lands at the start of a time series, exactly where an emergence detector is most sensitive. We implement a temporal guard that withholds knowledge-base links predating a document, but the harder version, which disambiguates senses against a contemporaneous corpus, is unattempted.

Three further directions. *Per-occurrence stance* with a trained classifier rather than a cue lexicon, which the current gold set cannot evaluate. *Successor relations*: co-occurrence and role transitions

already suggest when one technology displaces another, and extracting explicit comparative statements (“outperforms X”) would turn that into a graph (Hope et al., 2021). *Full-text patents*: the description section carries the specific disclosure that claim language deliberately generalises away, and extending the zone there is mostly a cost-control problem given the batching in §3.7. We would also like to compare against a fine-tuned span tagger (Beltagy et al., 2019; Zhong and Chen, 2021) trained on a larger gold set, and against domain extractors in chemistry (Krallinger et al., 2015) on the patent subset, neither of which the present annotation budget supported.

10 Conclusion

Technology extraction for evolution tracking is a precision-dominant, open-vocabulary problem in which fabricated entities do lasting damage and missed ones mostly do not. Treating that asymmetry as a design constraint — offsets rather than strings, cited evidence rather than assertion, one calibrated abstention rather than many cautious stages — produces a system whose hallucination rate is zero by construction and whose remaining errors are boundary and type disagreements that a reader can inspect. The components that matter most are not the ones that find mentions but the ones that decide which to keep.

Reproducibility

Corpora are rebuilt from public endpoints without credentials; all offline numbers are produced by make data kb gold eval. Each result row is a configuration object rather than a code path, every mention records the digest of the configuration that produced it, and model calls are cached content-addressed so a run replays exactly. The tables in this paper are generated from the results file and the numbers quoted in prose are LaTeX macros from the same source.

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